

FT85BS Manual V1.5

Specifications :

- Firmware: V1.5 or above (firmware update via Flipsky ESC Tool supported)
- Voltage: 14-85V(safe for 4-20S)
- Continuous current: 50V/60A; 75V/50A; 85V/40A (Values depend on the mounting, ambient temperature and air/water circulation around the device!)
- Peak Current: 250A
- Communication Ports: USB, CAN, UART, PPM, ADC
- Input Signals Support: PPM, ADC, PAS, UART(compatible with Flipsky remote such as VX1, VX1Pro, VX2, VX3, VX3Pro, VX4Pro)
- Startup Modes Support: IPDS (sensorless), HFPI (sensorless), HALL (sensored), ENCODER(ABI Encoder, AS5047 Absolute Type Encoder SPI Port), OPEN (Sensorless Open-Loop Startup)
- Traction Control: YES
- BEC: 5V@1A
- Multiple Speed Gear levels: Low, Medium, High, Reverse
- Programmable: YES
- Regenerative Capacity: YES
- Motor/power wire size: 12 AWG
- Size: L110.5*W55*H30mm(Including Alu. Case)

Features:

- Adopting ARM Cortex-M4 structure with the max 240MHz , improves PWM switching frequency and the motor RPM.
- Programmable via Flipsky ESC Tool to be compatible with multi-types of motor, different battery voltages and many control ways.
- Programmable via Flipsky remote controller VX4Pro (remote firmware version: VX4Pro_V1.1 or above).
- Supporting motor FOC with low working noise and smooth starting.
- Supporting Real-time data and fault alerts.
- Supporting adaptive variable PWM frequency control.
- Supporting sensor mode(HALL, ENCODER) and sensorless mode(IPDS, HFPI, OPEN).
- Supporting control types: Current, Duty Cycle, Speed control, Position Control
- Protection functions: low and high voltage protection, over-current protection, temperature abnormally protection, MOSFET over-temperature and over current protection, battery over discharge protecting, motor over-temperature protection.
- Supporting multiple typed of motor temperature sensors such as NTC-10K-25℃ , NTC-10K-100℃ , PT1000, PTC-1K-100℃ , KTY and so on.

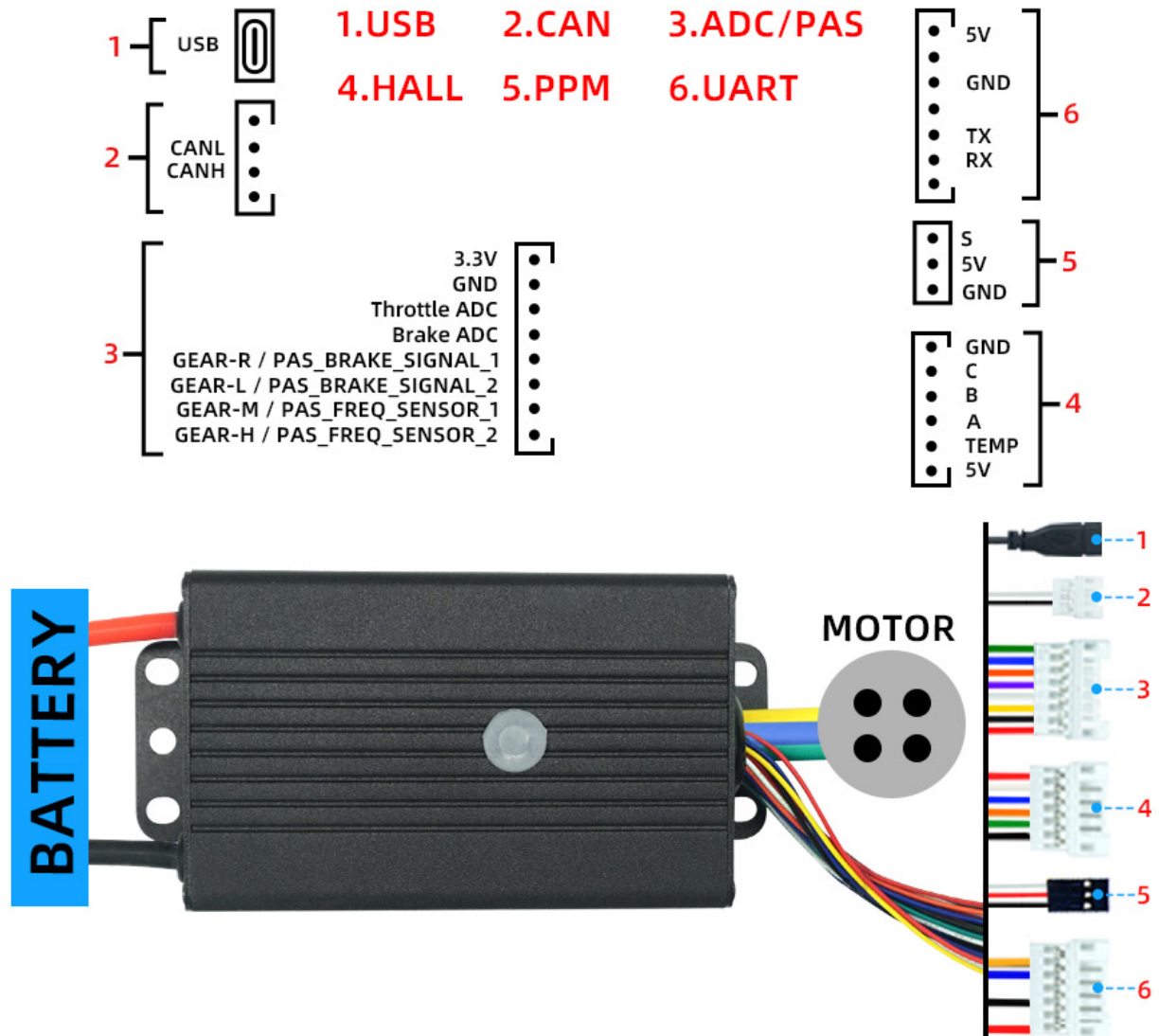
<https://flipsky.net/>

E-mail:flipskytech@outlook.com

Application Fields:

Electric skateboard / Ebike / E-Scooter / Electric Motorcycle / Robotics / FPV / RC Toys / Boat / Kayak etc.

Wiring diagram:



To download the Flipsky ESC Tool and User Manual, please scan the QR code:



<https://flipsky.net/>

E-mail:flipskytech@outlook.com

FT85BS规格说明书（版本V1.5）

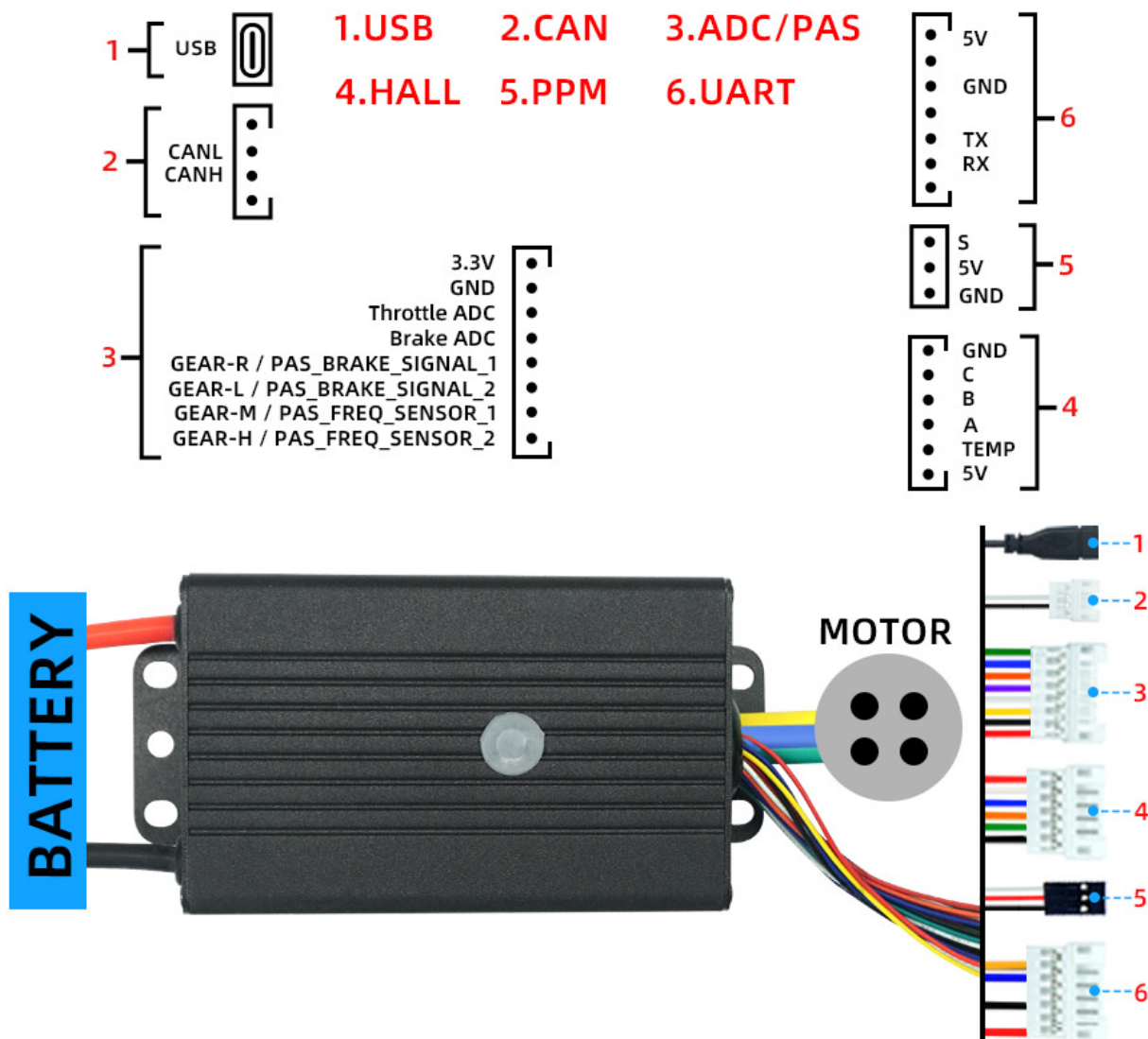
规格参数：

- 固件：V1.5及以上，支持通过Flipsky ESC Tool升级
- 电压：14V-85V (安全范围：4S-20S)
- 持续电流：50V/60A、75V/50A、85V/40A（取决于电调安装位置以及设备环境的温度、湿度等）
- 峰值电流：250A
- BEC:5V@1A
- 调参工具：Flipsky ESC TOOL
- PWM变频控制：YES
- 可再生制动：有
- 通讯端口：USB，CAN，UART，PPM，ADC
- 控制信号类型：PPM，ADC，PAS, UART（目前兼容 VX1、VX1Pro、VX2、VX3、VX3Pro、VX4Pro 等遥控器）
- 启动模式：IPDS(无感轻载)，HFPI(无感重载)，HALL，ENCODER（ABI编码器，AS5047绝对式编码器SPI接口），OPEN(无感开环启动)
- 牵引力控制：YES
- 速度档位：低速档，中速档，高速档，倒车档
- 线缆：12AWG (电机&电源)
- 尺寸：110.5*55*30mm（带外壳）

产品特点：

- 采用ARM Cortex-M4架构，高达240MHz主频，可提高PWM控制频率和电机转速上限
- 可通过 Flipsky ESC Tool 编程，兼容多种电机、不同电池电压和多种控制方式；
- 支持 Flipsky VX4Pro 遥控器调参（固件版本：VX4Pro_V1.1 以上版本）；
- 支持 FOC，工作噪音低，启动平稳；
- 支持实时数据和故障报警支持自适应PWM变频控制
- 支持传感器模式（HALL、ENCODER）和无传感器模式（IPDS、HFPI、OPEN）
- 支持控制类型：电流（Current），占空比（Duty Cycle），速度控制（Speed Control），位置控制（Position Control）
- 保护功能：低压保护；高压保护；电机过温保护；MOS管过流保护；电池过放保护；MOS管过温保护；
- 支持多种类型电机温度传感器类型，包括NTC-10K-25°C、NTC-10K-100°C、PT1000、PTC-1K-100°C、KTY系列等多种类型温度传感器

接线图：



Flipsky ESC TOOL 软件+中文用户手册下载二维码：

